



MAT1320-Linear Algebra

Lecture Notes

Vectors

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Vectors

Vectors: Physical Point of View

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- On the other hand, there are also quantities, such as force and velocity, that possess both **magnitude** and **direction**.
- These quantities, which can be represented by arrows having appropriate lengths and directions and emanating from some given reference point A , are called **vectors**.

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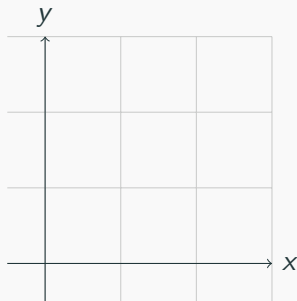
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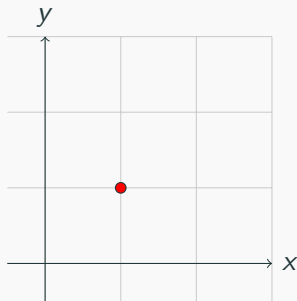


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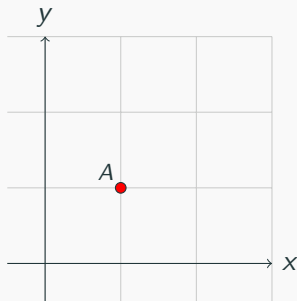


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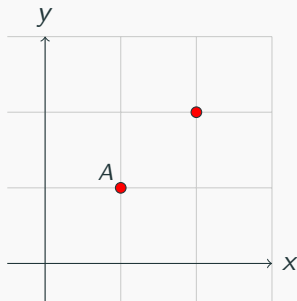


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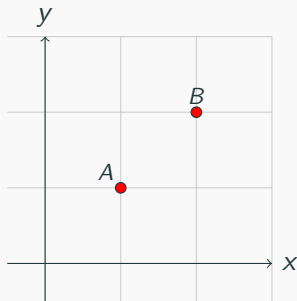


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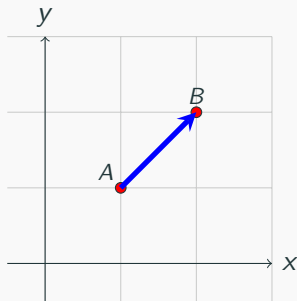


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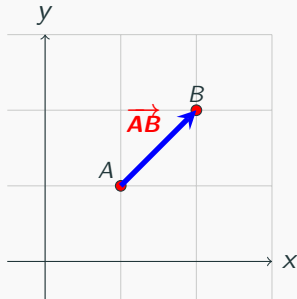


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Norm (Length) of a Vector

- Let $P_0(x_0, y_0, z_0)$ and $P_1(x_1, y_1, z_1)$ be two points in $\mathbb{R}^3$. Then the vector with start point P_0 and end point P_1 is denoted by $\overrightarrow{P_0P_1}$ and defined as

$$\vec{v} = \overrightarrow{P_0P_1} = (x_1 - x_0, y_1 - y_0, z_1 - z_0).$$

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$P_0(1, -2)$ and $P_1(-1, 3)$ be two points in $\mathbb{R}^2$.

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Note: For any nonzero vector $\vec{v}$, the vector $\frac{\vec{v}}{|\vec{v}|}$ is the unique unit vector in the same direction as $\vec{v}$.

Vector Addition

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$$\vec{u} + \vec{v} = (u_1 + v_1, u_2 + v_2, \dots, u_n + v_n)$$

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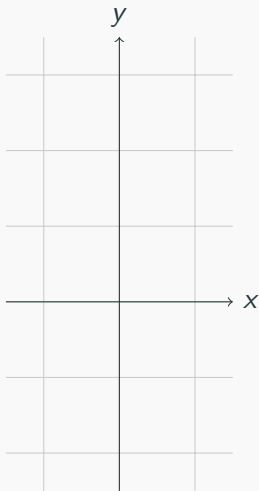
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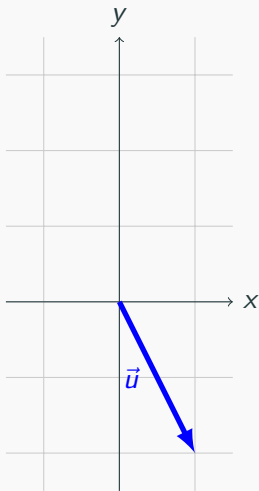


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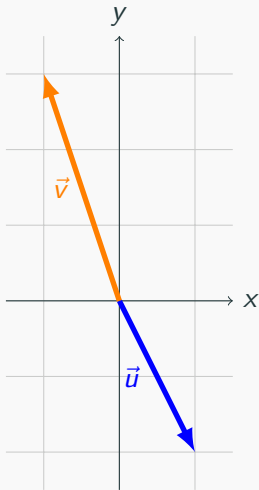


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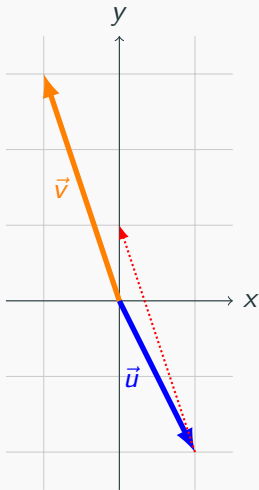


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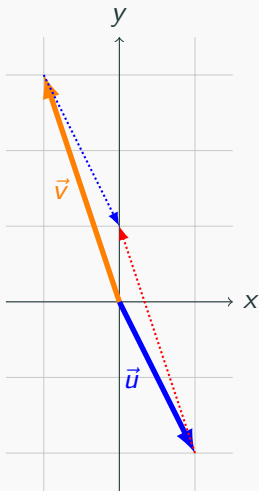


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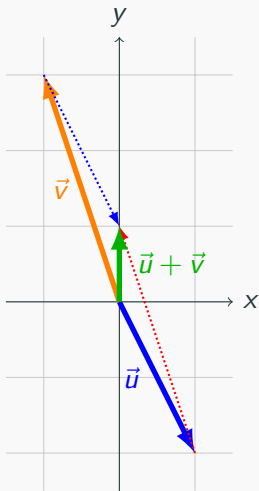


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4. If $\lambda < -1$ and $1 < \lambda$, then $|\vec{u}| < |\lambda \vec{u}|$.

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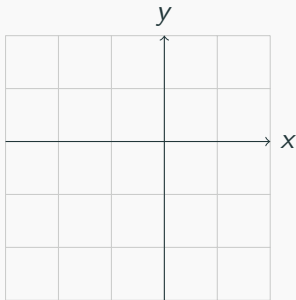
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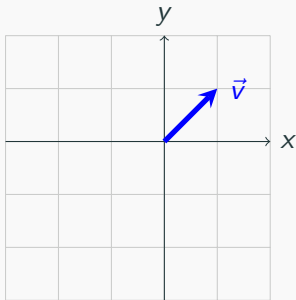
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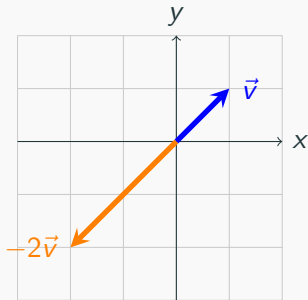
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Example

If $\vec{\mathbf{u}} = (2, 1, 0, 1)$ and $\vec{\mathbf{v}} = (-1, 1, 3, 2)$, then the dot product of $\vec{\mathbf{u}}$ and $\vec{\mathbf{v}}$ is

$$\begin{aligned}\vec{\mathbf{u}} \cdot \vec{\mathbf{v}} &= u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4 = \sum_{i=1}^4 u_i v_i \\ &= 2(-1) + 1 \cdot 1 + 0 \cdot 3 + 1 \cdot 2 = 1.\end{aligned}$$

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Note: The dot product of two vectors is a real number.

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Let the vectors $\vec{u} = (u_1, u_2, \dots, u_n)$, $\vec{v} = (v_1, v_2, \dots, v_n)$, $\vec{w} = (w_1, w_2, \dots, w_n)$ the scalars $\alpha, \beta \in \mathbb{R}$ be given. The inner product satisfies the following properties:

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6. $|\vec{u} \cdot \vec{v}| \leq |\vec{u}| |\vec{v}|$ (Schwartz Inequality)

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4. $\vec{0} \cdot \vec{u} = 0$
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7. $\vec{u} \cdot \vec{v} = |\vec{u}| |\vec{v}| \cos \theta$

Dot (Inner) Product: Properties

Let the vectors $\vec{u} = (u_1, u_2, \dots, u_n)$, $\vec{v} = (v_1, v_2, \dots, v_n)$, $\vec{w} = (w_1, w_2, \dots, w_n)$ the scalars $\alpha, \beta \in \mathbb{R}$ be given. The inner product satisfies the following properties:

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The Angle Between Two Nonzero Vectors

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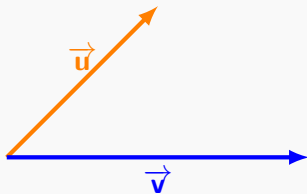
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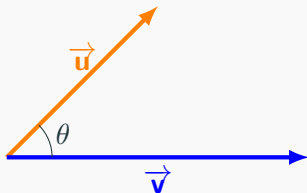
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Find the angle between the vectors $\vec{u} = (2, 2, 0, 1)$ and $\vec{v} = (-1, 1, 0, 2)$.

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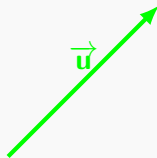
Projection of A Vector onto Another

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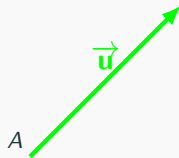


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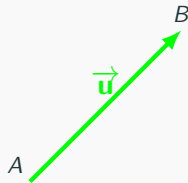


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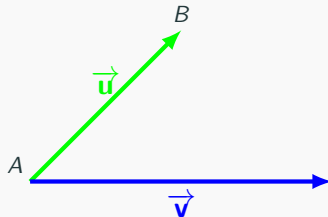


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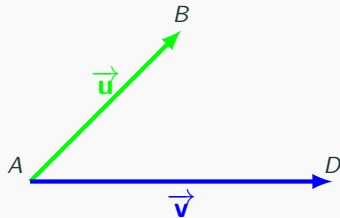


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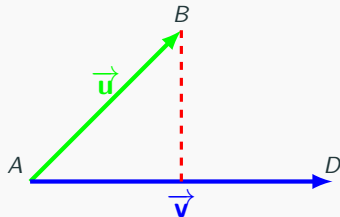


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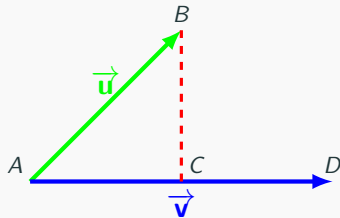


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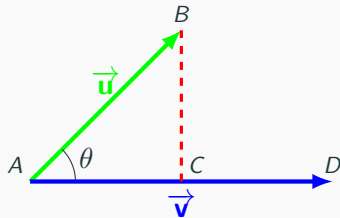


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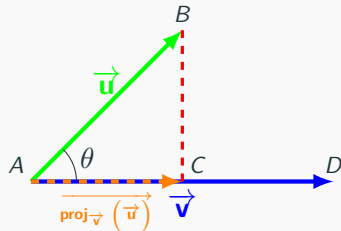


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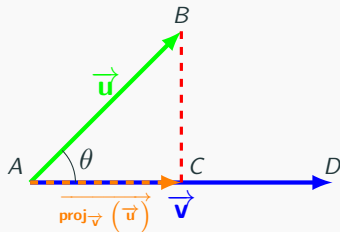


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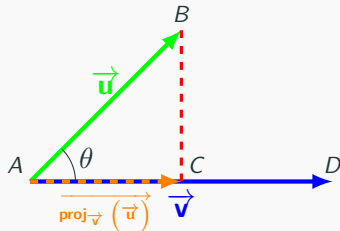
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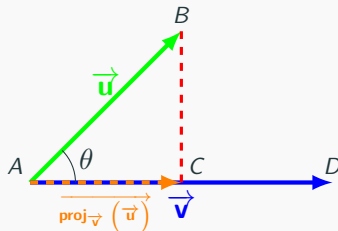
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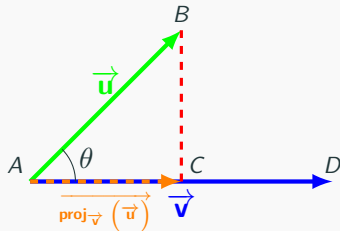


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Find $\text{proj}_{\vec{v}}(\vec{u})$.

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$$\frac{\vec{u} \cdot \vec{v}}{|\vec{v}|} \cdot \frac{\vec{v}}{|\vec{v}|}$$

$$-\frac{1}{3}, \frac{1}{3}, 0, \frac{2}{3}$$

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$$\frac{-2+2+2}{\sqrt{6}} \cdot \frac{1}{\sqrt{6}}$$

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Definition

Let $\vec{u} = (u_1, u_2, u_3)$ and $\vec{v} = (v_1, v_2, v_3)$ be two vectors in $\mathbb{R}^3$. The **cross product** of the vectors $\vec{u}$ and $\vec{v}$ is denoted by $\vec{u} \times \vec{v}$ or $\vec{u} \wedge \vec{v}$ and defined as follows:

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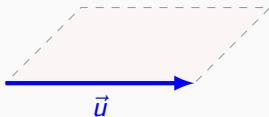
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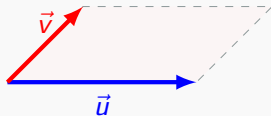


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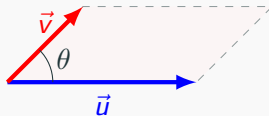


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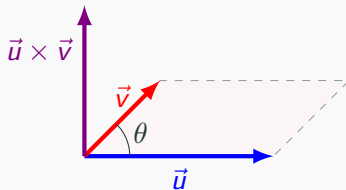


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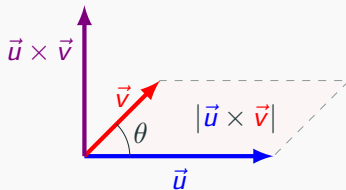


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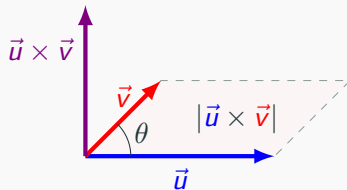
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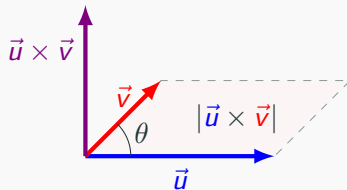


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Cross Product

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Cross Product

Example

Let $\vec{u} = (1, 2, -1)$ and $\vec{v} = (-2, 3, 4) \in \mathbb{R}^3$. Find the cross product of $\vec{u}$ and $\vec{v}$.

$$\begin{vmatrix} 1 & 2 & -1 \\ -2 & 3 & 4 \end{vmatrix}$$

$$(8+3)\mathbf{i} - (4-2)\mathbf{j} + (3+4)\mathbf{k}$$

$$11\mathbf{i} - 2\mathbf{j} + 7\mathbf{k}$$

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Mixed Product

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Definition

Let $\vec{u} = (u_1, u_2, u_3)$, $\vec{v} = (v_1, v_2, v_3)$ and $\vec{w} = (w_1, w_2, w_3)$ be three vectors in $\mathbb{R}^3$. Then the **mixed product** of the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ is denoted by $\vec{u} \cdot (\vec{v} \times \vec{w})$ or $(\vec{u}, \vec{v}, \vec{w})$

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Note: Mixed product is only defined over $\mathbb{R}^3$.

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Geometrically, mixed product of the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ is the volume of the parallelepiped having edges as $\vec{u}$, $\vec{v}$ and $\vec{w}$.

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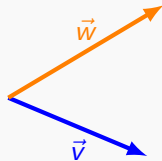


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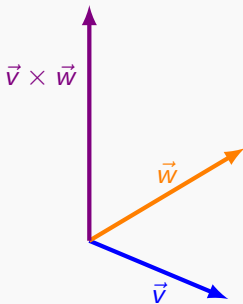


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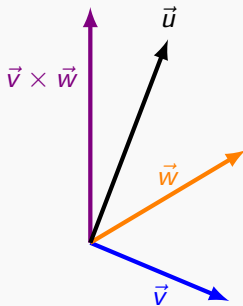


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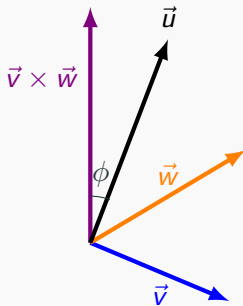


Mixed Product: Geometric Interpretation

Geometrically, mixed product of the vectors $\vec{u}$, $\vec{v}$ and $\vec{w}$ is the volume of the parallelepiped having edges as $\vec{u}$, $\vec{v}$ and $\vec{w}$.

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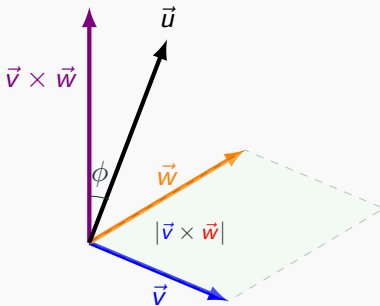
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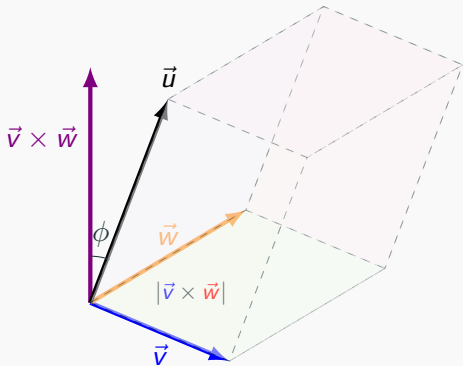
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Mixed Product

Example

The mixed product $\vec{u} \cdot (\vec{v} \times \vec{w})$, of the vectors $\vec{u} = (1, 2, -1)$, $\vec{v} = (-2, 3, 4)$ and $\vec{w} = (2, 1, 0) \in \mathbb{R}^3$ is

$$\vec{u} \cdot (\vec{v} \times \vec{w}) = \begin{vmatrix} 1 & 2 & -1 \\ -2 & 3 & 4 \\ 2 & 1 & 0 \end{vmatrix}$$

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Mixed Product: Properties

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Let $\vec{\mathbf{u}} = (u_1, u_2, u_3)$, $\vec{\mathbf{v}} = (v_1, v_2, v_3)$, $\vec{\mathbf{w}} = (w_1, w_2, w_3)$ and $\vec{\mathbf{r}} = (r_1, r_2, r_3) \in \mathbb{R}^3$ $c \in \mathbb{R}$.

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5. $\vec{u} \cdot (\vec{v} \times \vec{w}) = 0 \Leftrightarrow \vec{u} \parallel \vec{v} \text{ or } \vec{u} \parallel \vec{w} \text{ or } \vec{v} \parallel \vec{w} \text{ or } \vec{u} \parallel \vec{v} \parallel \vec{w}$

Two Fold Cross Product

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The **two fold cross product** of the vectors $\vec{u} = (u_1, u_2, u_3)$, $\vec{v} = (v_1, v_2, v_3)$ and $\vec{w} = (w_1, w_2, w_3) \in \mathbb{R}^3$ is defined by

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Note: The result of two fold cross product is a vector over $\mathbb{R}^3$.

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Example

Let $\vec{u} = (1, 2, -1)$, $\vec{v} = (-2, 3, 4)$ and $\vec{w} = (2, 1, 0) \in \mathbb{R}^3$ given. Find $\vec{u} \times (\vec{v} \times \vec{w})$.

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$$4 \quad (-8, 12, 16) -$$

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$$\vec{u} \cdot \vec{w} = 1 \cdot 2 + 2 \cdot 1 + (-1) \cdot 0 = 4$$

$$\vec{u} \cdot \vec{v} = 1 \cdot (-2) + 2 \cdot 3 + (-1) \cdot 4 = 0$$

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?